CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

Enthusiast COURSE

PHASE : SPEED & ACCURACY TEST

TARGET : PRE-MEDICAL 2022

Test Type :**PRACTICE TEST** Test Pattern :**NEET (UG)**

TEST DATE : ________

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	2	2	1	2	3	4	4	1	2	3	4	2	1	1	2	4	1	4	4	4	2	4	3	4	1	1	3	3	3

Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A.	4	1	2	4	1	3	1	1	1	4	4	1	3	4	4

HINT – SHEET

1. Ans (2)

We know that

$e_{in} = B_Z l_Y v_X$

$e_{in} = 5 \times 5 \times 1 = 25V$

2. Ans (2)

As the angle between field and plane is 60°, therefore angle between area vector and magnetic field is,

$\theta = 30^\circ$

Magnetic flux through the strip is

$\phi = BA\cos\theta$

$10^{-3} = B(0.02)\cos 30^\circ$

$$B = \frac{10^{-3}}{(0.02)\left(\frac{\sqrt{3}}{2}\right)}$$

$B = 0.058T$

3. Ans (2)

Induced current is

$$i = \frac{1}{R}\left(N\frac{d\phi}{dt}\right) = \frac{1}{R}\left(NA\frac{dB}{dt}\right)$$

$$i = \frac{1}{20}(10)(10 \times 10^{-4})(10^4) = 5A$$

$i = 5A$

4. Ans (1)

Both coils oppose the increase in current in straight wire, due to which change in magnetic flux is increased through the two coils. Hence current in coil A is clockwise and current in coil B is anticlockwise.

5. Ans (2)

As we have to consider the length perpendicular to velocity vector and magnetic field vector independent of shape and length of wire, the perpendicular length between two wires is constant hence value of induced current will remain constant

6. Ans (3)

$$E = -\frac{d\phi}{dt}$$

$$E = -\frac{d}{dt}(5t^2 - 6t + 9)$$

$E = 6 - 10t$

$t = 0s, E(0) = 6 - 0 = 6V$

$t = 0.5s, E(0.5) = 6 - 5 = 1V$

$$\frac{E(0)}{E(0.5)} = \frac{6}{1}$$

7. Ans (4)

Magnitude of self inductance is

$$L = \frac{e}{\frac{di}{dt}}$$

$$L = \frac{5}{\frac{(3-2)}{10^{-3}}} = 5 \times 10^{-3}$$

$L = 5mH$

8. Ans (4)

The magnetic field inside a toroid is,

$$B = \frac{\mu_0 Ni}{2\pi R}$$

Flux linked with the toroid,

$$\phi = \pi b^2 \times B \times N$$

Also, flux is related to self-inductance as,

$$\phi = Li$$

$$L = \frac{\phi}{i} = \frac{\mu_0 N^2 b^2}{2R} \text{ with } b <<< R$$

$$\text{Energy} = \frac{1}{2}Li^2 = \frac{\mu_0 N^2 I^2 b^2}{4R}$$

9. Ans (1)

Induced EMF is,

$$E = -N\frac{\Delta\phi}{\Delta t}$$

$$E = \frac{-NBA(\cos\theta_2 - \cos\theta_1)}{\Delta t}$$

$$E = -\frac{NBA(\cos 180^\circ - \cos^\circ 0)}{\Delta t}$$

$$E = \frac{2NBA}{t}$$

$$E = \frac{2(200)(0.3)(70\times10^{-4})}{0.1}$$

$E = 8.4V$

10. Ans (2)

$$e = M\frac{di}{dt} = 0.005 \times \frac{d}{dt}(I_0 \sin\omega t)$$

$$= 0.005 \times I_0 \omega\cos\omega t$$

$$\therefore e_{max} = 0.005 \times 10 \times 100\pi = 5\pi$$

$[\because \cos\omega t = 1]$

.

11. Ans (3)

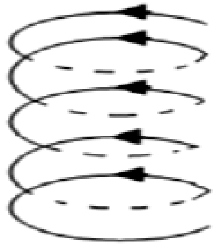
When an aluminum plate is placed near to the coil, electromagnetic induction in the aluminum plate gives rise to the electromagnetic damping that helps to stop the motion.

12. Ans (4)

$$L = -\frac{e}{\frac{di}{dt}} = \frac{-8 \times 0.05}{-4} = 0.1H$$

13. Ans (2)

The spring will compress because the force of attraction between two adjacent turn carrying current in the same direction. Will compress it.



14. Ans (1)

From

$$\frac{dI}{dt} = \frac{e}{L} = \frac{90}{0.2} = 450As^{-1}$$

15. Ans (1)

Here, $I = i_0$, at $t = \infty$

Let i be the current at $t = 1$ s

$$\text{From } i = i_0\left(1 - e^{-\frac{R}{L}t}\right)$$

$$i = i_0\left(1 - e^{-\frac{10}{5}\times 1}\right) = i_0\left(1 - \frac{1}{e^2}\right)$$

$$\therefore \frac{i_0}{i} = \frac{e^2}{e^2 - 1}$$

16. Ans (2)

$e = B(v\sin 30^o)\, l = 4$ volt

By fleming right hand, I_{ind} will be B to A

$V_A - V_B = 4V$

17. Ans (4)

On closing the switch, current begins flowing through coil 1. Hence , a magnetic field will be produced in this coil. The magnetic field varies from zero to a maximum value in a certain time. As a result, the filed across coil 2 will change for that time. This change in field will cause current to be induced in coil 2 for that short time. This phenomenon of generation of current in coil2 caused by the variation in the magnetic field of coil 1, is known as the electromagnetic induction. The current induced in coil 2 will deflect the pointer of the galvanometer connected to it.

Why alternate A is wrong: coil 2 will not move towards coil 1 because no mechanical force is produced on closing the switch.

Why alternate B is wrong: coil 2 will not move away from coil 1 because there is no mechanical force that could repel it.

Why alternate C is wrong: since current will be induced in coil 2 the pointer of the galvanometer will not remain at zero.

18. Ans (1)

$\phi = \vec{B} \cdot \vec{A}$

The area vector $\vec{A}$ is perpendicular to the surface, So when the surface is parallel to the magnetic field, the angle between $\vec{B}$ and $\vec{A}$ is 90^{o}

Hence, $\phi = 0$

19. Ans (4)

$$\frac{V_s}{V_p} = \frac{I_p}{I_s}$$

$$\text{OR } I_s = I_p \times \frac{V_p}{V_s} = 5 \times \frac{220}{22000} = 0.05\text{A}$$

20. Ans (4)

$e = 200 \sin 100\pi t$

We have

$e_0 = 200$

$\omega = 100\pi$

$BAN\omega = e_0$

$$B = \frac{200}{(0.25 \times 0.25) \times 1000 \times 100\pi}$$

$B = 0.01$ T

21. Ans (4)

Initial flux linked with inner coil when $i = 0$ is zero.

Final flux linked with inner coil when $i = i$ is

$$\phi = \left(\frac{\mu_0 i}{2b}\right) \pi a^2$$

$$|\Delta q| = \left|\frac{\Delta \phi}{R}\right|$$

$$\Delta q = \left(\frac{\mu_0 i}{2b}\right) \frac{\pi a^2}{R}$$

22. Ans (2)

Here $l = 50$ m, $v = 360$ km/h$=100$ m/s

$B = 2\times10^{-4} \text{wbm}^{-2}$

Potential difference

$e = Blv = 2\times10^{-4}\times50\times100 = 1V$

23. Ans (4)

As electron approaches the loop flux linked with the loop first increases and then decreases as electron passes by. Therefore direction of induced current changes.

24. Ans (3)

Potential difference across coil is

$$V = L\frac{di}{dt} = (2)(4) = 8V$$

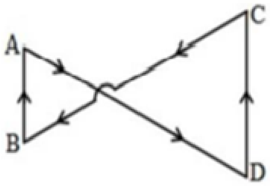
Now Energy stored per unit time

Power $= Vi = (8)(2) = 16$ J/s

25. **Ans (4)**

In the given question the magnetic field directed into the paper is increasing at a constant rate.

∴ Induced current should produce a magnetic field directed out of the paper. Thus current in both loops must be anticlockwise considering individually.



Induced emf on right side of loop will be more as area of its loop on right side is more.

$$\left[\because e = -\frac{d\varphi}{dt} = -A\frac{db}{dt}\right]$$

$e \propto A$

Hence the net current induced in the complete loop will be along DCBAD.

26. **Ans (1)**

$$\frac{80}{100} = \frac{120 \times 20}{1000 \times I_p}$$

$$I_p = \frac{120 \times 20}{1000 \times 0.8} = 3A$$

27. **Ans (1)**

Equivalent resistance is 8 Ω Therefore time constant is,

$$\tau = \frac{L}{R_{eq}}$$

$$\tau = \frac{2}{8} = 0.25s$$

And the steady state current is,

$$i_0 = \frac{\varepsilon}{R}$$

$$i_0 = \frac{6}{8} = 0.75A$$

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28. **Ans (3)**

$$\varepsilon_L = \varepsilon_0 e^{-\frac{t}{\tau}}$$

When current is allowed to flow through coil. Therefore the graph is

29. **Ans (3)**

In steady state , both inductors will be short circuit and current drawn from the cell is

$$I_0 = \frac{V_0}{R}$$

This current will be divided in L_1 and L_2 like in parallel resistors

$$I_1 = I_0 \frac{L_2}{(L_1 + L_2)}$$

30. **Ans (3)**

Given self-inductance, L=$1.8\times10^{-4}H$

Resistance , R=6Ω

When self-inductance and resistance are broken up into identical coils.

Then, the self-inductance of each coil

$$= \frac{1.8}{2} \times 10^{-4}H$$

The resistance of each coil $= \frac{6}{2} = 3\Omega$

Coil are then connected in parallel

$$L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$$

$$L' = \frac{\frac{1.8}{2} \times 10^{-4} \times \frac{1.8}{2} \times 10^{-4}}{\frac{1.8}{2} \times 10^{-4} + \frac{1.8}{2} \times 10^{-4}} = 0.45\times10^{-4}H$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

$$\text{Time constant} = \frac{L'}{R'} = \frac{0.45}{1.5} \times 10^{-4}$$

$$= 0.3 \times 10^{-4}s$$

2^{nd} : $\tau_i = \frac{L}{R}$

$$L' = -\frac{L}{2}$$

$$R' = \frac{R}{2},$$

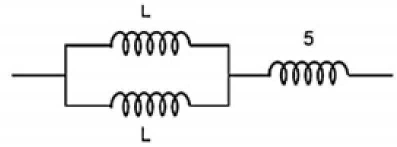
$$\tau'_{final} = \frac{L'}{R'} = \frac{L}{R}$$

$\tau' = \tau = 0.3 \times 10^{-4}S$

31. **Ans (4)**

$$\frac{L}{2}+5=15$$

$L = 20mH$



32. **Ans (1)**

As$M \propto N_1 N_2$

Therefore, M remians the same

33. **Ans (2)**

$$\text{Power output} = 3kW \times \frac{90}{100} = 2.7kW$$

$$I_s = 6A$$

$$\therefore V_s = \frac{2.7kW}{6A} = 450V$$

$$\Rightarrow I_p = \frac{3kW}{200V} = 15A$$

34. **Ans (4)**

Induced emf is,

$$\left[\because e = -\frac{d\varphi}{dt} = -A\frac{dB}{dt}\right]$$

Therefore induced emf and direction of current does not depend on material of the coil but resistance of copper is less than iron therefore current in copper coil is more.

35. **Ans (1)**

$$e = \frac{1}{2}B\omega l^2$$

$$e = \frac{1}{2}B\omega(2a)^2 = 2B\omega a^2$$

36. **Ans (3)**

$$\varepsilon = \frac{B\omega R^2}{2}$$

$$= \frac{0.1 \times (2\pi \times 20)(0.1)^2}{2}$$

$$= 20\pi \times 10^{-3}\text{volt}$$

37. **Ans (1)**

$$i = \frac{\varepsilon}{R}\left(1 - e^{-\frac{tR}{L}}\right)$$

Emf across is $\varepsilon_R = \varepsilon\left(1 - e^{-\frac{tR}{L}}\right)$

$$\Rightarrow \varepsilon_R = 6V\left(1 - e^{-\ln\sqrt{2}\cdot 2}\right)$$

$$\Rightarrow \varepsilon_R = 6V\left(1 - \tfrac{1}{2}\right) = 3V$$

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Voltage across coil

$\varepsilon_c = \varepsilon - -\varepsilon_R = 3\text{volt}$

38. **Ans (1)**

According to Lenz's Iaw, induced current will be developed in both the loops in opposite direction. Therefore, the net current in each loop will decrease.

39. **Ans (1)**

$$e = Bvl = 0.1\left(2 \times 10^{-2}\right) \times 6$$

$$= 12 \times 10^{-3} = 12mV$$

40. **Ans (4)**

$$|e| = \frac{d\phi}{dt}$$

$$= \frac{8 \times 10^{-4}}{0.4} = 2 \times 10^{-3}V$$

41. **Ans (4)**

Let

I = I_0sin < xclass =” symbol ”> ω < /x >t

Where $I_0 = 20A$, $\omega = 100\pi\text{rads}^{-1}$

Then,

$$\varepsilon = M\frac{dI}{dt} = M\left(\frac{d\,(I_0 \sin \omega t)}{dt}\right) = MI_0\omega \cos \omega t$$

$$\varepsilon = MI_0\omega = 10\pi$$

$$M = \frac{10\pi}{20 \times 100\pi} = 0.005H = 5mH$$

42. **Ans (1)**

If there is flux change $\Delta\phi$, then $q = \frac{N\Delta\emptyset}{R}$.

$q = \frac{1}{7} \times (1.35 - 0.79) = 0.08C$

43. **Ans (3)**

Induced emf,

$E = -\frac{d\emptyset}{dt}$

Therefore, E= 0, when $\frac{d\emptyset}{dt} = 0$ and vice-versa.

Even if $\varphi = 0$, $\frac{d\emptyset}{dt}$ may be non-zero.

Similarly,if $\varphi \neq 0$, $\frac{d\emptyset}{dt}$ may be zero.

For example, if $\phi = \phi_0 \sin\omega t$

then $E = -\frac{d\emptyset}{dt} = -\emptyset_0 \cos \omega t$

44. **Ans (4)**

In a transformer frequency remains same for input & output.

45. **Ans (4)**

Mutual inductance between coils is

$M = K\sqrt{L_1 L_2}$

$M = 1\sqrt{2 \times 10^{-3} \times 8 \times 10^{-3}}$

(K=1)

$m=4\times10^{-3}mH$